

Takahiro Matsunaga

Curriculum Vitae

Personal Information

Name: Takahiro Matsunaga

Affiliation: Department of Civil Engineering, the University of Tokyo

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Education

Ph.D. in Engineering

Since Oct, 2025

Behavior in Networks Studies Unit (Supervisor: Eiji Hato)

Department of Civil Engineering, the University of Tokyo

M.S. in Engineering (Outstanding Paper Award)

Apr, 2024 – Sep, 2025

Behavior in Networks Studies Unit (Supervisor: Eiji Hato)

Department of Civil Engineering, the University of Tokyo

B.S. in Engineering (Best Paper Award)

Apr, 2020 – Mar, 2024

Behavior in Networks Studies Unit (Supervisor: Eiji Hato)

Department of Civil Engineering, the University of Tokyo

Peer-Reviewed International Journal Papers

- [1] Takahiro Matsunaga and Eiji Hato. “Mean field dynamics for pedestrian interaction in quantized observation-based 3D route choice modeling”. In: *Transportation Research Part C: Emerging Technologies* 186 (2026), p. 105608. ISSN: 0968-090X. DOI: <https://doi.org/10.1016/j.trc.2026.105608>. URL: <https://www.sciencedirect.com/science/article/pii/S0968090X26000963>.

Peer-Reviewed National Journal Papers (in Japanese)

- [1] 松永隆宏 and 羽藤英二. “言語生成モデルに基づく首都圏移動活動経路の予測”. In: **都市計画論文集** 60.3 (2025), pp. 1527–1534.
- [2] 松永 隆宏 and 羽藤 英二. “BLE 観測の不確実性を考慮した屋内 3 次元経路選択モデルの推定”. In: **都市計画論文集** 59.3 (2024), pp. 1683–1690.

Peer-Reviewed International Conferences

- [1] Takahiro Matsunaga and Eiji Hato. “Dynamic Congestion Control in Metropolitan Railways using Generative Language Models: Approach by Bid-free Reverse Auctions”. In: *Transportation Research Board Annual Meeting (TRB 2025)*. Peer-reviewed. Washington, D.C., USA, 2026.

- [2] Takahiro Matsunaga and Eiji Hato. “Partial Controllability in Mean Field Incentive Dynamics”. In: *European Association for Research in Transportation (hEART 2026)*. Peer-reviewed. Paris., France, 2026.
- [3] Takahiro Matsunaga and Eiji Hato. “Activity Generation using Panel Data of Day-to-day Activity-travel Pattern in Tokyo Metropolitan Area”. In: *The 29th International Conference of Hong Kong Society for Transportation Studies (HKSTS 2025)*. Peer-reviewed, oral presentation. Hong Kong, 2025.
- [4] Takahiro Matsunaga and Eiji Hato. “Interaction of 3D pedestrian flow in a congested railway station: Structural estimation based on Mean Field Game theory”. In: *The 12th International Conference on Pedestrian and Evacuation Dynamics (PED 2025)*. Peer-reviewed, accepted, oral presentation. Prague, Czech Republic, 2025.
- [5] Takahiro Matsunaga and Eiji Hato. “Joint Estimation of Latent OD Matrix and 3D Route Choice Model Across Railway Station Boundaries based on Manifold Learning”. In: *The 28th International Conference of Hong Kong Society for Transportation Studies (HKSTS 2024)*. Peer-reviewed, oral presentation. Hong Kong, 2024.
- [6] Takahiro Matsunaga and Eiji Hato. “Three-Dimensional Pedestrian Route Choice Model with Quantization of Observational Uncertainty”. In: *IEEE International Conference on Intelligent Transportation Systems (ITSC)*. Peer-reviewed, oral presentation. Edmonton, Canada, 2024.
- [7] Takahiro Matsunaga and Eiji Hato. “Three-Dimensional Pedestrian Route Choice Model with Quantization of Observational Uncertainty”. In: *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE. 2024, pp. 1–8.

Oral Presentation without Review (in Japanese)

- [1] 松永 隆宏 and 羽藤 英二. “部分観測型交通需要の平均場制御–Causal Transformer による動学的選好の導入–”. In: **土木学会全国大会**. 2026.
- [2] Takahiro Matsunaga and Eiji Hato. “Incentive Design for Congestion Control of Public Transport Demands based on Reverse Auction”. In: **土木計画学研究発表会**. 2025.
- [3] 松永 隆宏 and 羽藤 英二. “複雑な day-to-day 活動動態学習に基づく首都圏アクティビティ生成モデル”. In: **交通工学研究発表会**. 2025.
- [4] 松永 隆宏 and 羽藤 英二. “駅構内における三次元経路選択モデルの多様体推定”. In: **土木学会全国大会**. 2024.
- [5] 鈴木 健史, 齋藤 功次, 山崎 淳, 本田 聖一朗, 羽藤 英二, 石井 良治, and 松永 隆宏. “渋谷駅山手線線路切替工事時における多様な観測データを用いた行動変容の実態分析”. In: **土木計画学研究・講演集 (CD-ROM)** 68 (2023), pp. 27–1.

Awards

- [1] *Outstanding Master Thesis*. The Department of Civil Engineering, The University of Tokyo, 2025.
- [2] *Outstanding Research Presentation*. Japan Society of Traffic Engineers, 2025.
- [3] *Outstanding Student Paper Award*. Hong Kong Society for Transportation Studies (HKSTS), 2025.
- [4] *Best Graduation Thesis (Dean’s Award)*. The Department of Civil Engineering, The University of Tokyo, 2023.

Fellowships

- [1] *JSPS Research Fellow (DC1)*. Japan Society for the Promotion of Science, Apr, 2026 -.

- [2] *SPRING GX (Support for doctoral student)*. The University of Tokyo, Fostering Advanced Human Resources to Green Transformation, Oct, 2025 -.
- [3] *WINGS-GLAFS (Support for master and doctoral student)*. The University of Tokyo, Institute of Gerontology, Oct, 2024 - Sep, 2025.

Other Public Outputs (in Japanese)

- [1] 羽藤 英二, 小川 大智, 古橋 郁一, 松永 隆宏, 今村 啓太, and 須賀 理帆. **現代行動理論の基礎と応用**. Behavior in Networks Studies Unit, The University of Tokyo. 2026. URL: <https://github.com/bin-utokyo/stone-ship>.